Data Engineer

Case Study

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Case Study - General Guidelines

Dear candidate.

This case study is divided into two parts:
- Part 1: focused on ETL creation
- Part 2: your capability to extract information from data

Following the stack used at Ebury, you must use only Python and SQL.
You will have 1 week to complete this case study.
We'll have a call for you to present your solution, so be prepared to do it.

Deliverable
Your main deliverable must be a .zip folder with:
- python code that solves the Part 1 and Part 2 (you can have one or multiple python files)
- a dockerfile that should install all dependencies and process your code
- subfolder used as volume to your dockerfile (your database should be here)
- a document detailing your answers, include any relevant artifacts such as diagrams, code snippets, text, images, etc.

Important
Make sure to document how to process your code. Considering add a step by step to facilitate the execution.

AI Usage
The use of AI is allowed but you must describe exactly how you are using it. AI tools are part of our stack, and in case of you use it, your method will be evaluated as well.

Please do not hesitate to contact us with any further questions.

Good luck! =)

Part 1 - ETL

You received a list of JSONs and you must create an ETL to ingest this data into our Data Lake.

This ETL must split the JSON data into the tables:

- Payment
- Partner

You might use SQLite as your database to simulates the Data Lake, so you can materialize the tables with the results.
You must create a volume on your docker outside the image so we can access your database without running the container.

Have in mind that your code will be executed multiple times.

Explicitly in your code the data contract.

Make sure your code is readable and well documented.

Part 2 - Querying

Considering the tables you create on the previous step, answer the following questions:

1. Count the number of transactions from each partner.
2. Calculate the minimum and the maximum exchangeRate.
3. Determine the number of trades per day.
4. Determine the foreignGrossAmount per currency
5. Determine the average cappingRate per currency

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